Siyuan Chen

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Education

- **Tsinghua University (THU)** Beijing, China
  Bachelor of Engineering in Computer Science and Technology *Expected Jul. 2027*

  Cumulative GPA: 3.84/4.0

  Selected Coursework:
  CS Core: Introduction to Artificial Intelligence, Introduction to Deep Learning, Data Mining, Data Structures
  Math: Probability & Statistics, Discrete Mathematics, Linear Algebra, Calculus

Research Interests

My research interests center on AI Safety and Alignment, with a particular focus on AI deception, trustworthy LLMs, LLM agents, and post-training alignment.

Publications

- Yao Huang*, **Siyuan Chen***, Yinpeng Dong. "**Why Static Probes Fail: Context Determines Deception Feature Composition**." *Submitted to ICLR 2027.* (* Equal Contribution)
- Wenhao Li, Ziran Wang, **Siyuan Chen**, et al. "**TACIT: From Circuit Discovery to Testable Algorithmic Explanations**." *Submitted to ACL Rolling Review 2026.*

Research Experience

- **LLM Agent Red Teaming and Safety Evaluation** University of Illinois Urbana-Champaign
  Visiting Scholar; Advisor: ***Prof. Bo Li*** *Jun. 2026 – Sep. 2026*
  - **Automated Red Teaming Pipeline**: Co-developed an automated pipeline to generate adversarial scenarios and systematically evaluate LLM agents across diverse safety categories, including **Critical Unauthorized Data Loss** and **Intellectual Property**.
  - **Agent Runtime and Evaluation Framework**: Co-developed a modular runtime and evaluation framework supporting multi-step tool invocation, interaction with MCP-backed simulated application environments (e.g., Gmail, Slack, and CRM systems), and standardized, reproducible execution for scalable LLM-agent red teaming.
  - **Large-scale Agent Safety Evaluation**: Conducted large-scale empirical evaluations of frontier LLM agents using the developed pipeline and runtime framework, analyzing safety performance across risk categories, application domains, adversarial injection vectors, and tool-use environments.
- **AI Deception Detection & Robustness Research** T-STAR Lab, Tsinghua University
  Research Assistant; Advisor: ***Prof. Yinpeng Dong*** *Sep. 2025 – May 2026*
  - **Geometry of Deception Analysis**: Characterized deception as a **context-dependent multi-dimensional subspace**, demonstrating the limitations of static linear probes for deception detection.
  - **Iterative Orthogonal Subspace Decomposition (IOSD)**: Proposed **IOSD** to sequentially extract orthogonal deception subspaces, enabling hierarchical feature extraction beyond single-direction representations.
  - **Dynamic Deception Subspace Navigation (DDSN)**: Developed **DDSN** with a **Domain-Adaptive Gating Mechanism**, achieving **0.96 AUROC** with only **10-shot** test-time adaptation.
- **Mechanistic Interpretability of Transformers** THUNLP, Tsinghua University
  Research Assistant; Advisor: ***Prof. Maosong Sun*** *Feb. 2025 – Aug. 2025*
  - **Testable Algorithmic Explanations**: Co-developed **TACIT**, an agentic framework that turns sparse task circuits into testable, evidence-grounded algorithmic explanations.
  - **Evidence-Guided Hypothesis Testing**: Designed LLM-agent workflows to generate and refine component hypotheses using attention, causal-intervention, and direct-output evidence.

Honors & Awards

2025 **Tsinghua Friends – Toyota Scholarship** *Awarded for Academic Excellence (Top 15%)*
2023 – 2025 **Freshman Scholarship** *Awarded to the 2023 cohort and renewed annually.*

Skills

- **Computer Skills**: Python, PyTorch, C++, Git
- **English Proficiency**: **GRE**: 331 (Verbal 161, Quantitative 170); **TOEFL**: 105 (Reading 29, Listening 30, Speaking 23, Writing 23)